

QORLA Stability Test 2

Trainable Q/K/V Dynamics, Optimizer-Aware Stabilization, Deep-Stack Failure, and Forward-Exact Correction Isolation

QORLA Project — Experimental Research Note

Abstract

This document records the complete second stability campaign for QORLA, beginning from the forward-preserving conditioning rule retained after Stability Test 1 and ending with a substantially different mechanistic conclusion. The campaign moved from trainable query/key/value projections to sequence-length generalization, conditioning-score redesign, correction-branch gradient decomposition, Adam-preconditioned update analysis, transient optimizer-aware compression, structured operator transfer, real supervised classification, deep stacked architectures, optimizer robustness, causal backward-path ablation, untouched multi-split validation, and a final runtime/memory audit. Several attractive intermediate mechanisms failed under broader validation. In particular, sequence-normalized conditioning scores did not produce a universal trigger; leverage-based event prediction was weak; scalar raw-gradient suppression was largely normalized away by Adam; optimizer-space correction dominance stabilized synthetic updates but over-regularized normal real-data training; and minibatch-local descent gates failed in deep stacks. The decisive deep-model finding was instead causal: propagating gradients through the local-linear correction term R is a major depth-amplified failure mode. Replacing the correction in the backward graph by $\text{sg}(R)$ while preserving the exact forward value R produced a monotonic dose response, substantially outperformed lower learning rate and global clipping, transferred across Adam, AdamW, and RMSprop, and passed a frozen 45-pair blind validation across three unseen train/test splits and depths $L \in \{1, 2, 4\}$. The final numerical audit measured exactly zero forward discrepancy, zero invalid gradient steps, about 29.9% lower step time, and about 5.6% lower peak allocated GPU memory in the tested implementation. Stability Test 2 therefore concludes with a depth-dependent, forward-exact correction stop-gradient rule and a refined mechanistic claim: *deep-QORLA collapse arises substantially from backward coupling through the correction pathway, not merely from large key gradients or generic optimizer instability.*

1 Purpose and scope

Stability Test 1 established that altering the forward local-linear correction can damage extrapolative behavior even when numerical conditioning improves. It therefore retained a forward-preserving backward damping rule based on an operator-level conditioning score. Stability Test 2 asks the harder question: *does that principle remain valid once queries, keys, values, upstream representations, optimizers, and multiple QORLA blocks are all trainable?*

The campaign deliberately treats failed ideas as evidence rather than deleting them from the research record. This is important because the final mechanism was not obtained by a single hyperparameter sweep. It emerged after progressively falsifying four increasingly broad hypotheses: (i) low operator conditioning alone explains trainable instability, (ii) correction dominance alone identifies harmful updates, (iii) optimizer-preconditioned correction dominance is a universal control signal, and (iv) local descent tests are sufficient in deep stacks. The final conclusion was reached only after causal branch ablation and frozen blind validation.

The main supervised test bed is the scikit-learn handwritten digits dataset: 1797 grayscale 8×8 images, represented as sequences of length $N = 64$, with embedding width $d = 32$, batch size 64, 15–20 training epochs depending on the experiment, Adam learning rate 3×10^{-3} unless otherwise stated, Laplace scale $\lambda = 4$, ridge $\rho = 10^{-3}$, and $\epsilon = 10^{-8}$. Earlier synthetic and structured-transfer experiments sweep $N \in \{32, 64, 96, 128, 192\}$ and use smaller controlled models.

Stability Test 2 research progression

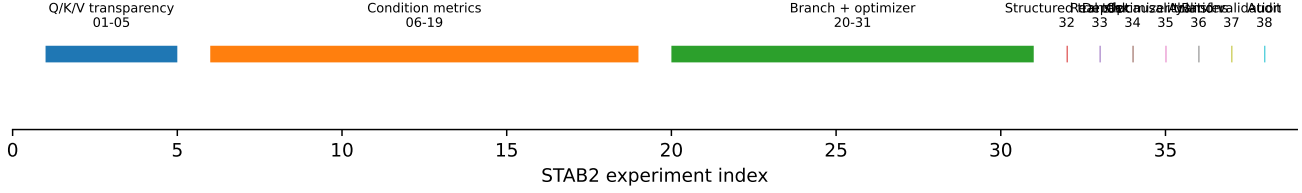


Figure 1: Research progression of Stability Test 2. The campaign spans trainable Q/K/V transparency, conditioning redesign, branch decomposition, optimizer-aware control, real-data transfer, deep-stack causality, optimizer robustness, final ablations, blind validation, and numerical auditing.

2 QORLA forward operator

For projected query coordinate a_i and key coordinate b_j , define

$$z_{ij} = b_j - a_i, \quad W_{ij} = \exp(-\lambda|z_{ij}|). \quad (1)$$

Let

$$S_{0,i} = \sum_j W_{ij}, \quad \mu_i = \frac{\sum_j W_{ij} z_{ij}}{S_{0,i} + \epsilon}, \quad (2)$$

$$Q_i = \sum_j W_{ij} (z_{ij} - \mu_i)^2, \quad C_i = \sum_j W_{ij} (z_{ij} - \mu_i) v_j. \quad (3)$$

The kernel-average base and local-linear correction are

$$B_i = \frac{\sum_j W_{ij} v_j}{S_{0,i} + \epsilon}, \quad (4)$$

$$R_i = -\mu_i \frac{C_i}{Q_i + \rho + \epsilon}, \quad (5)$$

$$o_i = \boxed{B_i + R_i}. \quad (6)$$

All retained Stability Test 2 mechanisms preserve this forward value exactly. The research question is entirely about how the computational graph carrying gradients through R should behave during training.

3 Initial hypothesis inherited from Stability Test 1

Stability Test 1 used

$$\kappa_i^{\text{old}} = \frac{Q_i + \rho}{S_{2,i} + \rho + \epsilon} \quad (7)$$

and a forward-preserving correction damping rule

$$\tilde{R}_i = \text{sg}(R_i) + s_i(R_i - \text{sg}(R_i)), \quad s_i = \min\left(1, \frac{\text{sg}(\kappa_i)}{\tau_g}\right), \quad (8)$$

with provisional $\tau_g = 0.03$. Stability Test 2 begins by inserting this principle into a genuinely trainable Q/K/V setting rather than assuming that the operator-level threshold is universal.

4 STAB2-01–05: trainable Q/K/V and the conditioning-persistence effect

4.1 Healthy transparency: STAB2-01A

A tiny trainable model with $B = 8$, $N = 96$, query/key width 64, value width 32, and trainable W_Q, W_K, W_V was constructed. In healthy geometry the minimum observed condition score was approximately $0.0635 > 0.03$. Forward outputs and losses matched the undamped model, gradient cosine similarities were exactly or numerically indistinguishable from one, and the damping rule remained silent. This established the first requirement: a stabilization rule must be transparent when the trainable geometry is healthy.

4.2 Stressed geometry: STAB2-01B

Separated narrow query/key clouds were produced with biases near $+1.2$ and -1.2 and small projection weights. The minimum score fell to approximately 0.0231 and all samples lay below 0.03 . The rule reduced Q/K/V gradient magnitudes while retaining high direction fidelity (cosines above about 0.9989). Thus the Stability Test 1 mechanism could activate conditionally inside a trainable model without changing the forward pass.

4.3 Persistent damping: STAB2-02

Multi-step training exposed a failure not visible in single-step tests. If low conditioning triggers damping of Q/K gradients, then the model receives less geometric signal to escape the low-conditioning region. This creates a feedback loop:

$$\text{low } \kappa \Rightarrow \text{damped Q/K gradients} \Rightarrow \text{slow geometry recovery} \Rightarrow \text{continued low } \kappa. \quad (9)$$

We refer to this as the *conditioning-persistence effect*. Permanent damping therefore stabilizes immediate gradients but can trap the learned geometry.

4.4 Hard release and smooth release: STAB2-03–05

A hard early damping window followed by abrupt release allowed geometry recovery but produced a rebound spike in W_Q . A smooth linear release from $\tau_g = 0.03$ to zero over the first 100 steps avoided the discontinuity. Five-seed confirmation showed approximate loss parity with mean W_Q spike reduction near 60% and W_K reduction near 31%, although W_V gradients increased by roughly 31%. The result was useful locally but already suggested that a single conditioning threshold might not generalize.

5 STAB2-06–13: sequence length and redesign of the conditioning score

The old score was found to carry sequence-length dependence because the fixed ridge ρ is mixed with unnormalized moment sums. As N grows, the ratio changes even under matched geometry. Controlled experiments therefore separated three quantities:

$$\kappa_{\text{old}} = \frac{Q + \rho}{S_2 + \rho + \epsilon}, \quad (10)$$

$$\kappa_{\text{geom}} = \frac{Q}{S_2 + \epsilon}, \quad (11)$$

$$\bar{q} = \frac{Q}{S_0 + \epsilon}, \quad \bar{m}_2 = \frac{S_2}{S_0 + \epsilon}, \quad (12)$$

$$\kappa_{\text{mass}} = \frac{\bar{q} + \rho}{\bar{m}_2 + \rho + \epsilon}. \quad (13)$$

The mass-normalized score was approximately invariant to controlled changes in sequence length and discriminated healthy, moderate, and severe geometries better than the old score. However, fixed thresholds and online quantile thresholds still failed when the geometry evolved during learning. The conclusion was methodological: *normalizing a conditioning score can repair scaling contamination without making that score a universal intervention trigger*.

6 STAB2-14–19: leverage and event prediction

A leverage-like statistic

$$\ell = \frac{\mu^2}{\text{Var}_W(z) + \rho} \quad (14)$$

was evaluated as a predictor of future W_K spikes. Algebraically, the mass-normalized score and leverage are near-equivalents:

$$\kappa_{\text{mass}} \approx \frac{1}{1 + \ell}. \quad (15)$$

Conditioning/leverage showed moderate rank correlation with next-step key-gradient events at larger N , but event precision and recall were not strong enough to define a robust trigger. Thresholds that captured most spikes over-intervened; conservative thresholds missed too many events. This closed the conditioning-only path and motivated a direct mechanistic decomposition of the gradient.

7 STAB2-20–31: correction-branch decomposition and the optimizer problem

7.1 Exact branch decomposition: STAB2-20

For the key projection, decompose

$$g_B = \nabla_{W_K} B, \quad g_R = \nabla_{W_K} R, \quad g_T = g_B + g_R. \quad (16)$$

Define correction dominance

$$D = \frac{\|g_R\|}{\|g_B\| + \epsilon}. \quad (17)$$

Across sequence lengths the strongest mechanistic result was that total key-gradient spikes were strongly associated with correction dominance: Spearman correlations were approximately 0.70–0.78 in synthetic experiments, spike events had roughly 1.7–1.9 $\times$ the ordinary R/B gradient dominance, and the branch cosine was usually positive rather than canceling. The spike mechanism was therefore not destructive branch cancellation; it was correction amplification.

7.2 Branch-relative caps: STAB2-21–28

A correction cap

$$s = \min \left(1, \alpha \frac{\|g_B\|}{\|g_R\| + \epsilon} \right) \quad (18)$$

strongly reduced synthetic spikes at $\alpha = 12$, but the absolute scale of D changed with architecture and sequence length. Fixed α values failed to transfer to the full residual/FFN block. Relative-EMA dominance caps also failed, especially at $N = 32$. These failures showed that correction dominance is a strong *diagnostic* but not automatically a universal threshold variable.

7.3 Threshold-free power compression: STAB2-29–31C

A threshold-free family

$$s_p = \min(1, D^{-p}) \quad (19)$$

was tested. Whole-branch $p = 0.10$ looked attractive on development seeds but failed held-out robustness: mean loss increased, worst relative loss degradation exceeded 25%, and several gradient-tail cases did not improve. Restricting the surgery to W_K with $p = 0.20$ produced strong raw-gradient suppression, motivating a direct audit of the optimizer update.

7.4 Adam audit: STAB2-31D

The Adam audit changed the direction of the project. K-only surgery reduced raw gradient statistics substantially:

$$\max \text{ gradient ratio} \approx 0.712, \quad \text{P95 gradient ratio} \approx 0.674, \quad (20)$$

but the actual Adam update ratios were nearly unchanged:

$$\max \text{ update ratio} = 1.000, \quad \text{P95 update ratio} \approx 0.9935. \quad (21)$$

The reason is the approximate scale invariance of Adam. If $g \mapsto sg$, then the first moment scales as $m \mapsto sm$ and the second moment as $v \mapsto s^2v$, giving

$$\frac{m}{\sqrt{v}} \mapsto \frac{sm}{\sqrt{s^2v}} \approx \frac{m}{\sqrt{v}}. \quad (22)$$

This established a central principle for the rest of Stability Test 2: *stability must be evaluated in optimizer-update space, not merely gradient space.*

8 STAB2-31E–31L: Adam-preconditioned correction dominance and transient PCD

To operate after preconditioning, branch-specific first moments are tracked:

$$m_t^B = \beta_1 m_{t-1}^B + (1 - \beta_1)g_B, \quad (23)$$

$$m_t^R = \beta_1 m_{t-1}^R + (1 - \beta_1)g_R, \quad (24)$$

while the ordinary full-gradient second moment is

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)(g_B + g_R)^2. \quad (25)$$

After bias correction,

$$u_B = \frac{\hat{m}_B}{\sqrt{\hat{v}} + \epsilon_A}, \quad u_R = \frac{\hat{m}_R}{\sqrt{\hat{v}} + \epsilon_A}, \quad (26)$$

and define optimizer-space dominance

$$D_A = \frac{\|u_R\|}{\|u_B\| + \epsilon}. \quad (27)$$

Preconditioned correction dominance (PCD) uses

$$s(D_A) = \min(1, D_A^{-p}), \quad \Delta W_K = -\eta(u_B + s u_R). \quad (28)$$

Development experiments selected $p = 0.30$ as the strongest update-tail setting. On blind seeds 10–19, PCD produced 50/50 wins for maximum update and top-10% update, 49/50 for P95 update, with mean ratios 0.763, 0.860, and 0.887 respectively. Apparent very large relative loss errors at $N = 32$ were traced to tiny baseline losses; absolute differences were small.

A permanent PCD penalty harmed late convergence. A finite transient window performed better. The frozen transient schedule became

$$p_t = \begin{cases} p_0 \left(1 - \frac{t-1}{T_s-1}\right), & t \leq T_s, \\ 0, & t > T_s, \end{cases} \quad p_0 = 0.30, \quad T_s/T = 0.30. \quad (29)$$

STAB2-31K then tested this transient PCD (TPCD) on fresh seeds 20–29 and $N \in \{32, 64, 96, 128, 192\}$: all 50 runs improved maximum, P95, and top-10% optimizer-update tails. The geometric mean update ratios were approximately 0.758, 0.929, and 0.909. Seed-cluster bootstrap over ten independent seeds confirmed the effect, while convergence-tail statistics remained approximately at parity.

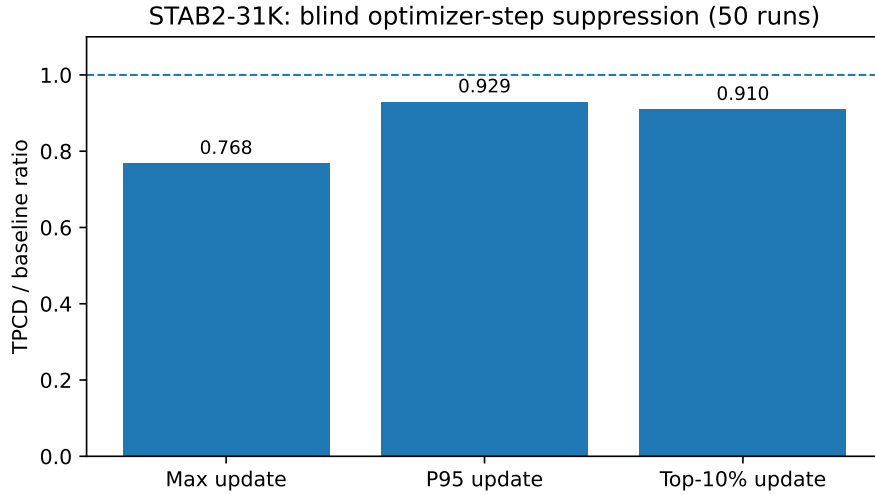


Figure 2: STAB2-31K. Frozen transient PCD strongly suppresses optimizer-step extremes across 50 blind synthetic runs while preserving convergence.

9 STAB2-32: structured learning transfer

STAB2-32A replaced static synthetic objectives by a matched QORLA teacher. Across 25 runs, TPCD achieved mean validation-loss ratio 0.9843, won 24/25 validation comparisons, and reduced maximum/P95/top-10% optimizer-update statistics to approximately 0.753/0.823/0.825 of baseline. STAB2-32B then used a mismatched Gaussian-kernel teacher; the student remained QORLA. Mean validation-loss ratio fell to 0.9194, all 25 validation comparisons improved, and all 25 runs improved each update-tail metric. These results established that optimizer-space stabilization could transfer beyond the synthetic objective used to derive it.

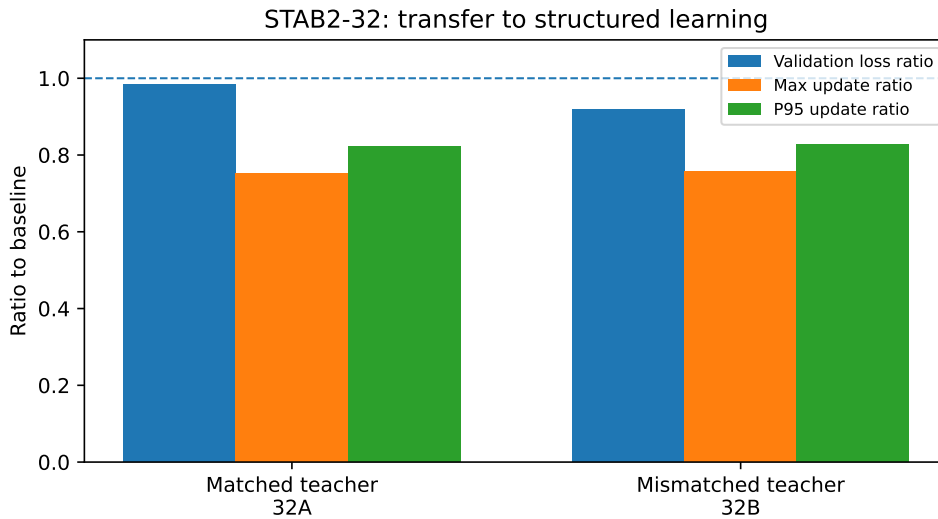


Figure 3: STAB2-32. TPCD transfers to matched and mismatched structured learning objectives, with strong optimizer-step suppression and no task degradation in these settings.

10 STAB2-33: real supervised digits and failure of dominance-only control

10.1 Stressed initialization: STAB2-33A

A supervised digits classifier used a single QORLA block, sequence length 64, width 32, batch size 64, and Adam at 3×10^{-3} . Under deliberately stressed Q/K initialization, old TPCD retained optimizer-tail benefits but task preservation was mixed: mean test-loss ratio was about 1.053 and mean accuracy change about -0.17 percentage points. This was a warning that optimizer stability need not imply task preservation.

10.2 Normal initialization: STAB2-33B

Under ordinary PyTorch initialization, dominance-only TPCD clearly failed as a universal real-data method. Mean test-loss ratio rose to 1.1515, mean accuracy changed by -0.83 percentage points, and one seed suffered a large task collapse even though its update magnitudes were heavily suppressed. The key inference was that large D_A can be normal and useful.

10.3 Correction utility diagnostic: STAB2-33C

Let the current raw key gradient be $g_K = g_B + g_R$. Define correction descent alignment

$$\chi = g_K^\top u_R \quad (30)$$

and normalized correction utility

$$U_R = \frac{g_K^\top u_R}{|g_K^\top (u_B + u_R)| + \epsilon}. \quad (31)$$

On baseline normal-init training, the correction was descent-positive about 79.9% of the time and median utility was approximately 0.949. In contrast, Spearman correlation of D_A with update magnitude and utility was near zero. Hence

$$\boxed{D_A \gg 1 \not\Rightarrow \text{harmful correction.}} \quad (32)$$

10.4 Descent-preserving TPCD: STAB2-33D–33E

A second-generation rule preserved the correction whenever it was locally descent-supporting:

$$s_t = \begin{cases} 1, & \chi_t > 0, \\ \min(1, D_A^{-p_t}), & \chi_t \leq 0. \end{cases} \quad (33)$$

On development seeds 45–54, this DP-TPCD restored task preservation: mean test-loss ratio 0.9794 and mean accuracy change $+0.69$ percentage points, while P95 and top-10% update ratios were about 0.926 and 0.944. The intervention was active only about 6.2% of steps. A blind confirmation on seeds 55–64 was interrupted by computer sleep after seed 63; seed 64 was rerun separately. The task-side aggregate from the available logs was approximately mean test-loss ratio 0.92 and mean accuracy change $+0.31$ percentage points, with no catastrophic collapse. Because complete update-tail bookkeeping for seeds 55–63 was not retained, STAB2-33E is recorded as a *provisional task-preservation pass*, not as a complete blind optimizer-tail confirmation.

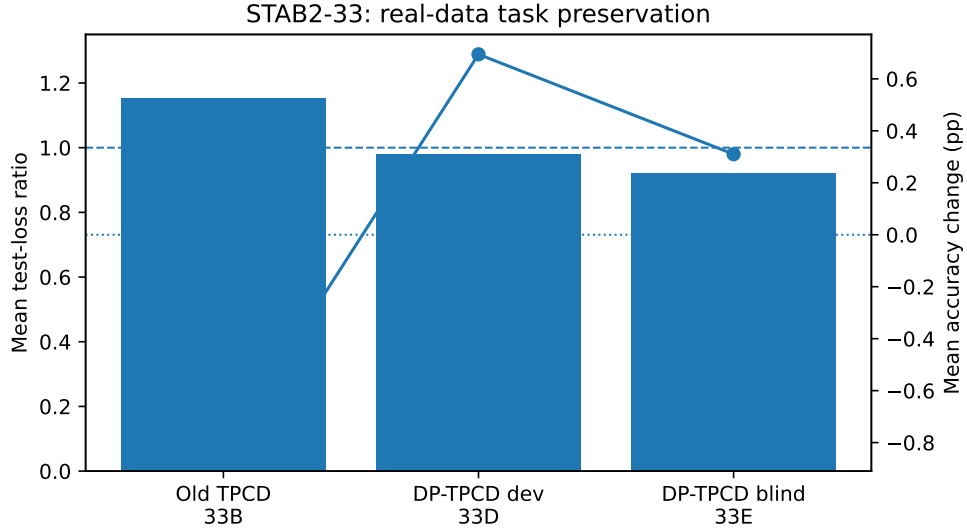


Figure 4: STAB2-33. Dominance-only TPCD fails under normal real-data initialization, while descent-preserving TPCD repairs task preservation on development and provisional blind runs.

11 STAB2-34: architecture depth exposes a different failure mechanism

11.1 Depth generalization failure: STAB2-34A

DP-TPCD was tested at $L \in \{1, 2, 4\}$ with seeds 65–69. It failed decisively: mean test-loss ratios were 1.227, 1.286, and 1.400 respectively, with mean accuracy changes -1.89 , $+0.44$, and -3.56 percentage points. P95 and top-10% update-tail wins were only 9/15 and 10/15. The worsening task effect despite lower average D_A at larger depth showed that dominance magnitude itself could not explain the deep failure.

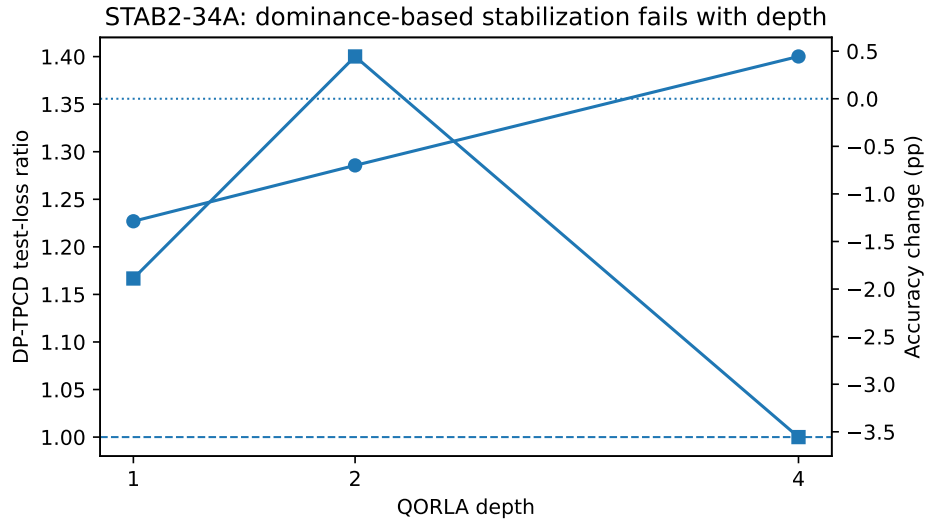


Figure 5: STAB2-34A. The real-data depth test falsifies the hypothesis that selective dominance control is a universal deep-QORLA stabilizer.

11.2 Projection, layer localization, and minibatch-consensus tests: STAB2-34B–34E

A mathematically minimal projection removed only the ascent component of u_R :

$$\tilde{u}_R = \begin{cases} u_R, & g^\top u_R \geq 0, \\ u_R - \frac{g^\top u_R}{\|g\|^2 + \epsilon} g, & g^\top u_R < 0. \end{cases} \quad (34)$$

This transient descent projection improved mean loss relative to the failed 34A rule but still degraded task performance and often worsened update tails. Applying it to one block at a time showed that all four blocks could hurt individually; no single layer explained the instability.

A cross-batch diagnostic then evaluated the same u_R against the current minibatch and an independent probe minibatch. Sign agreement was only about 68–71%, and approximately 39–44% of directions labeled harmful by the current batch were useful on the probe. A two-batch consensus gate was therefore tested, but it remained highly trajectory-dependent: some seeds improved dramatically while others collapsed. This falsified the hypothesis that minibatch noise alone explains the deep failure.

11.3 Trajectory sensitivity and early-warning diagnostics: STAB2-34F–34I

Keeping initialization fixed while changing only minibatch order produced large baseline variation. Across three initializations and four plans, mean baseline accuracy standard deviation was 6.39 percentage points and mean within-initialization range was 13.80 points. The consensus intervention increased those values to 8.51 and 18.80 points. Deep QORLA was therefore intrinsically trajectory-sensitive before stabilization.

Layerwise diagnostics suggested large early block-0 key gradients and correction/base forward energy ratio $\|R_0\|/\|B_0\|$ in bad development trajectories. However, a blind validation on new seeds 70–74 falsified the R/B hypothesis: correlations with final accuracy were essentially zero at 5, 10, 20, and 30 percent of training, and AUC values were close to chance. This prevented another predictor-tuning loop and motivated a direct causal ablation of the backward graph.

11.4 Causal branch ablation: STAB2-34J

Three forward-identical variants were compared at $L = 4$ on fresh seeds 75–79 and four batch orders:

- **BASE**: ordinary gradients through R ;
- **KSTOP**: stop only the $R \rightarrow W_K$ pathway;
- **ALLSTOP**: keep R in the forward pass but detach the complete correction branch during backward.

BASE averaged only 10.47% accuracy. KSTOP averaged 11.93% and produced only a small mean improvement. ALLSTOP averaged 66.83%, improved all 20 paired runs, and gained 56.36 percentage points on average. Therefore the deep failure is not primarily a key-gradient phenomenon. It is distributed through the full correction backward path into Q, K, V, and upstream representations.

11.5 Final depth confirmation: STAB2-34K

A final fresh test compared BASE, a 30% transient stop, and permanent RSTOPALL at $L \in \{1, 2, 4\}$ with seeds 80–84. Forward equality was exactly zero error. At $L = 2$, RSTOPALL improved mean accuracy from 24.06% to 74.28% and won 5/5. At $L = 4$, it improved 10.50% to 66.17% and won 5/5. Shallow $L = 1$ behavior remained variable, showing that the mechanism is depth-dependent rather than universally beneficial.



Figure 6: STAB2-34K. Permanent correction stop-gradient strongly rescues depth-2 and depth-4 QORLA training while shallow behavior remains more variable.

12 STAB2-35: optimizer robustness

12.1 Adam and AdamW: STAB2-35A

At $L = 4$, seeds 85–89, and four batch orders, RSTOPALL was compared under Adam and AdamW. Adam accuracy increased from 13.44% to 55.65% with 19/20 wins; AdamW increased from 12.57% to 57.33% with 20/20 wins. The number of runs below 20% accuracy fell from 18 to 0 for Adam and from 17 to 0 for AdamW. Thus the effect was not an artifact of decoupled versus coupled weight decay inside the Adam family.

12.2 SGD: STAB2-35B

Momentum SGD at learning rates 0.01, 0.03, and 0.10 was floor-limited: BASE accuracy remained near chance for almost all runs. RSTOPALL gave an overall +3.57 percentage-point trend and 16/30 wins, but because the baseline optimizer itself failed to train the model, the experiment could not establish or refute non-Adam transfer.

12.3 RMSprop: STAB2-35C

RMSprop provided the required non-Adam confirmation. At learning rates 10^{-3} and 3×10^{-3} , RSTOPALL improved mean accuracy from 12.61% to 48.36% and from 15.83% to 54.94%, respectively. All 20 paired comparisons improved, collapse counts fell to zero, and overall test loss was about $0.603 \times$ baseline. Therefore the correction-backward instability is not specific to Adam’s first-moment formulation.

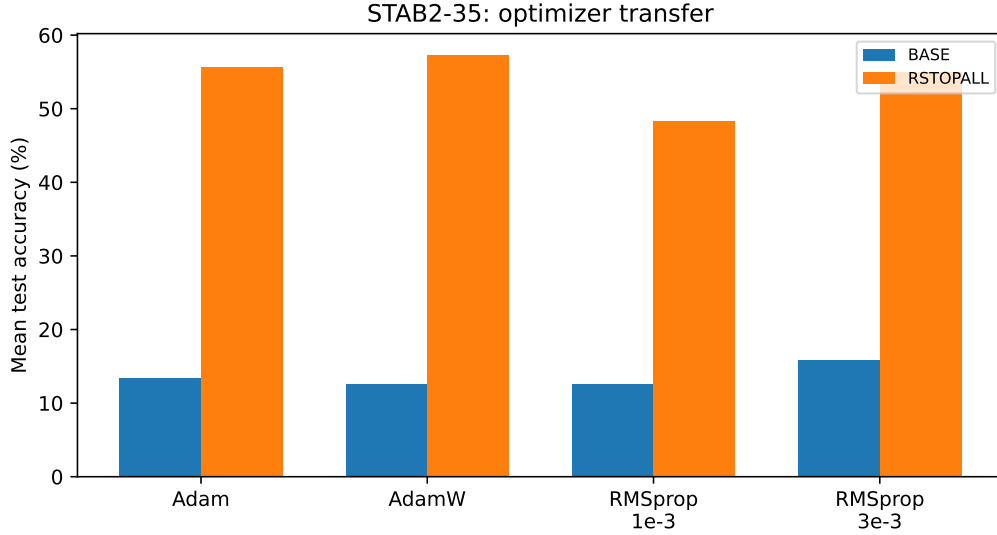


Figure 7: STAB2-35. Permanent correction stop-gradient transfers strongly across Adam, AdamW, and RMSprop.

13 STAB2-36: causal dose response and competing stabilization baselines

13.1 Backward-strength dose response: STAB2-36A

To test causality rather than a binary trick, define

$$R_\alpha = \text{sg}(R) + \alpha(R - \text{sg}(R)). \quad (35)$$

For every α , the forward value is exactly R , while

$$\frac{\partial R_\alpha}{\partial \theta} = \alpha \frac{\partial R}{\partial \theta}. \quad (36)$$

Fresh seeds 100–104 and three batch plans tested $\alpha \in \{1, 0.5, 0.25, 0\}$. Forward error was exactly zero for all settings. Mean accuracy increased monotonically as backward coupling was removed:

$$10.67\% (\alpha = 1) \rightarrow 14.06\% (0.5) \rightarrow 27.41\% (0.25) \rightarrow 59.69\% (0). \quad (37)$$

Collapse counts below 20% fell from 15/15 to 13/15, 5/15, and finally 0/15. Spearman correlation was

$$\rho(\alpha, \text{accuracy}) = -0.798, \quad \rho(\alpha, \text{test loss}) = +0.840. \quad (38)$$

At $\alpha = 0$, all 15 runs beat BASE and the worst paired gain was still +17.5 percentage points. This is the strongest causal evidence in the campaign.

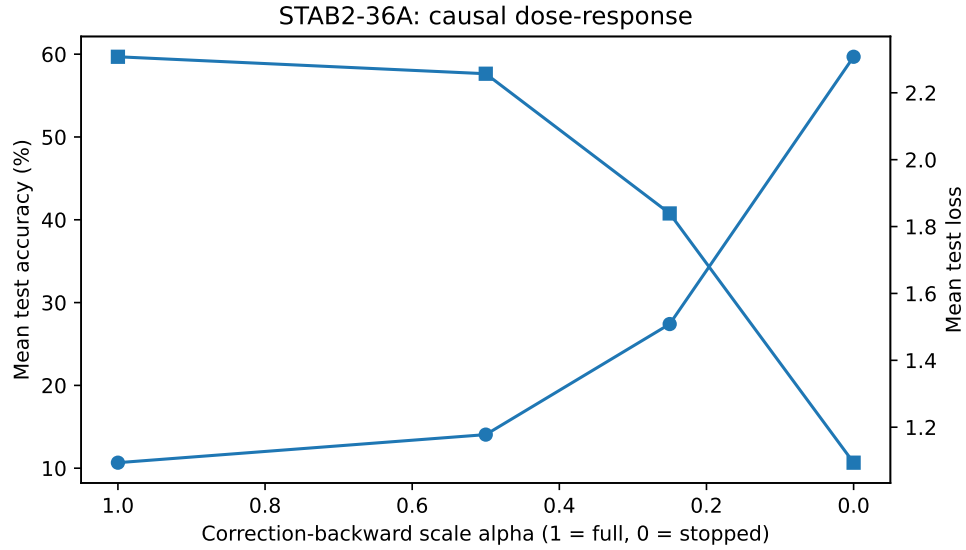


Figure 8: STAB2-36A. Exact-forward dose response: weakening only the correction backward pathway produces a strong monotonic improvement in deep-QORLA trainability.

13.2 Low learning rate and global clipping: STAB2-36B

A reviewer-style alternative explanation is that QORLA merely needs ordinary stabilization. Fresh seeds 105–109 and three batch plans therefore compared BASE, Adam with learning rate 10^{-3} , global gradient clipping at norm 1, and RSTOPALL. Mean accuracies were 12.61%, 16.72%, 16.15%, and 62.65%. RSTOPALL won all 15 paired comparisons with mean gain +50.04 percentage points and worst gain +20 points. Low learning rate won 10/15 with mean gain +4.11 points; clipping won 8/15 with mean gain +3.54 points. Clipping was active on about 34.9% of steps, so its weak effect was not caused by an inactive threshold. This establishes that targeted correction-backward isolation addresses a QORLA-specific failure mode not reproduced by generic update reduction.

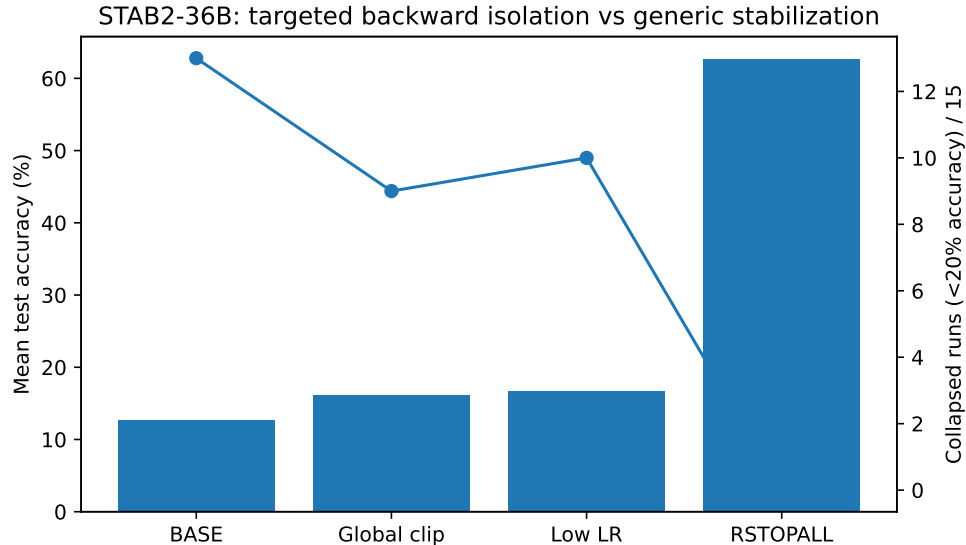


Figure 9: STAB2-36B. RSTOPALL strongly outperforms generic lower learning rate and global clipping in deep-QORLA collapse prevention.

14 STAB2-37: frozen untouched multi-split validation

After STAB2-36, the method was frozen: no new thresholds, alphas, triggers, or architecture-specific tuning. The final rule is simply

$$\tilde{R} = \text{sg}(R) \quad \text{in backward, while the forward value remains } R. \quad (39)$$

STAB2-37A tested three unseen stratified train/test split seeds 2027, 31415, and 92653; fresh model seeds 110–114; and depths $L \in \{1, 2, 4\}$. This produced 45 paired BASE/RSTOPALL comparisons.

Overall mean test-loss ratio was 0.4772, mean accuracy gain was +41.17 percentage points, and RSTOPALL won 42/45 pairs. By depth:

Depth	Mean loss ratio	Mean accuracy gain	Wins	Worst accuracy gain
$L = 1$	0.7086	+16.67 pp	12/15	-33.61 pp
$L = 2$	0.2919	+53.06 pp	15/15	+13.61 pp
$L = 4$	0.4310	+53.80 pp	15/15	+28.61 pp

Paired one-sided Wilcoxon tests gave $p = 0.00852$ for $L = 1$ and $p = 3.05 \times 10^{-5}$ for both $L = 2$ and $L = 4$. Each of the three unseen splits produced 14/15 wins. Thus the strongest claim is explicitly depth-focused: RSTOPALL robustly prevents the deep collapse at $L = 2$ and $L = 4$, while shallow behavior is positive in aggregate but more heterogeneous.

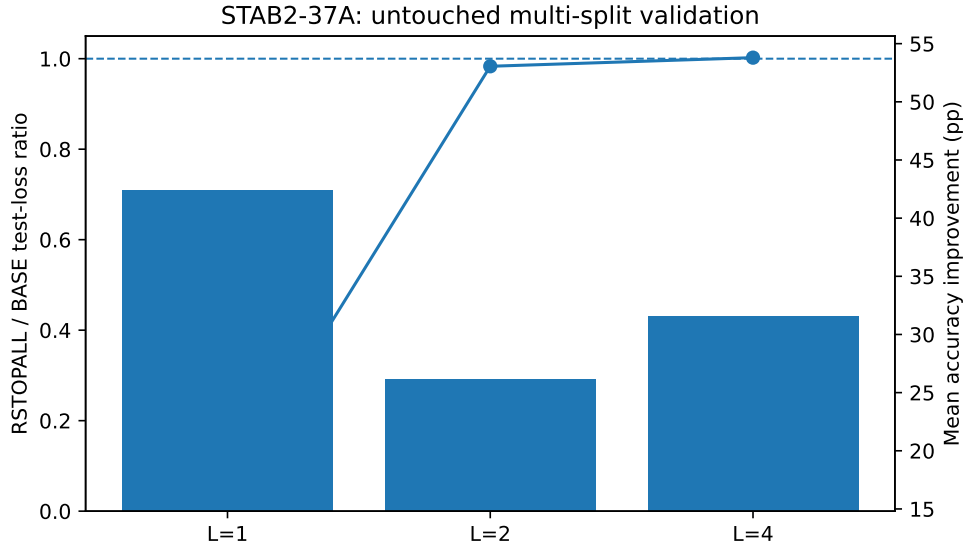


Figure 10: STAB2-37A. Frozen untouched multi-split validation. Deep settings show 30/30 paired wins and large accuracy gains.

15 STAB2-38: final numerical, runtime, and memory audit

The last audit compared ordinary backward propagation with RSTOPALL for $L \in \{1, 2, 4\}$ on GPU. Forward logits were compared directly and showed

$$\max |f_{\text{BASE}} - f_{\text{RSTOP}}| = 0.0 \quad (40)$$

at every tested depth. Both methods produced zero invalid (NaN/Inf) steps in the audit.

Measured step times were:

Depth	BASE ms/step	RSTOPALL ms/step	Ratio
1	6.261	4.113	0.6569
2	8.351	6.126	0.7336
4	15.255	10.881	0.7133

The mean training-time ratio was 0.7013, corresponding to about 29.9% lower step time in this implementation. Mean peak-memory ratio was 0.9441, or about 5.6% lower allocated peak GPU memory, with the memory advantage increasing from roughly 3.0% at $L = 1$ to 8.6% at $L = 4$. Gradient norms also fell, but the earlier clipping ablation prevents interpreting smaller norm alone as the cause of the task improvement.

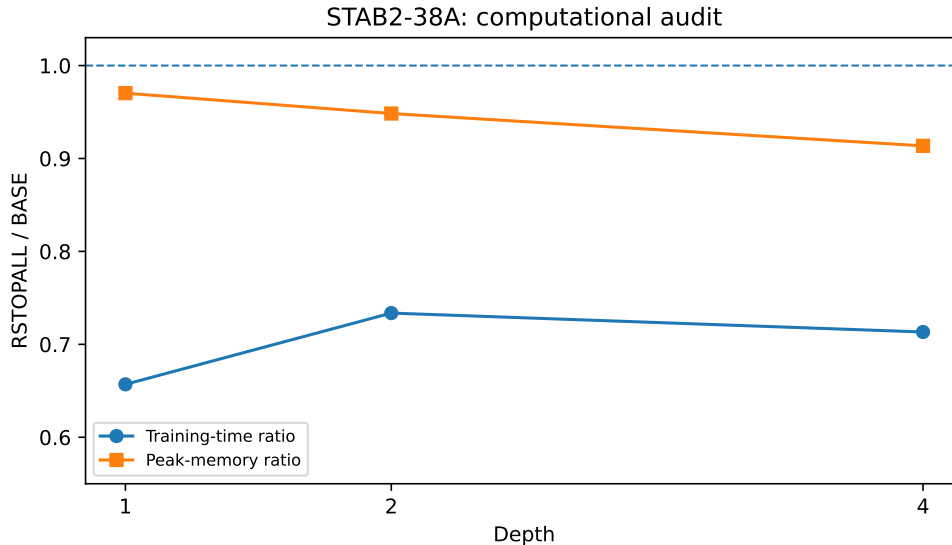


Figure 11: STAB2-38A. Forward-exact correction stop-gradient also reduces measured step time and peak allocated GPU memory because the correction branch is not retained for backward differentiation.

16 Final retained Stability Test 2 configuration

The retained forward operator is unchanged:

$$o_i = B_i + R_i. \quad (41)$$

During training, the correction term is represented as

$$R_i^{\text{train}} = \text{sg}(R_i) \quad (42)$$

for backward propagation. Equivalently, if an implementation requires the forward identity to be explicit,

$$R_i^{\text{train}} = \text{sg}(R_i) + 0 \cdot (R_i - \text{sg}(R_i)). \quad (43)$$

The base pathway B remains fully differentiable, and the forward numerical value of the QORLA layer is exactly the original operator. No conditioning threshold, dominance threshold, power exponent, or transient window is required by the final deep-model rule.

The final principle differs materially from the provisional conclusion of Stability Test 1:

In trainable deep QORLA stacks, the dominant failure mode is not simply poor local conditioning or excessive key-gradient magnitude. A major causal source of collapse is backward coupling through the local-linear correction branch itself. Preserve the exact forward correction, but isolate its backward pathway when deep-stack trainability is the objective.

17 Evidence hierarchy and defensible claims

The campaign supports the following claims at different strength levels.

Strongly supported

- Raw-gradient suppression can be almost canceled by Adam’s scale normalization; optimizer-step statistics are therefore essential.
- Correction-gradient dominance strongly diagnoses synthetic key-gradient spikes, but its absolute scale is not universal across architecture or sequence length.
- Dominance-only and local descent-based controls do not generalize reliably to normal real-data deep stacks.
- Deep QORLA training is highly sensitive to initialization and minibatch trajectory under ordinary backward propagation.
- Full correction-backward isolation is causally effective in deep stacks, while K-only isolation is not.
- The correction-backward strength exhibits a strong dose response: reducing the backward scale from 1 toward 0 monotonically improves deep trainability in the tested setting.
- The effect is not reproduced by generic lower learning rate or global gradient clipping.
- The effect transfers strongly across Adam, AdamW, and RMSprop.
- The frozen method passes untouched multi-split validation at $L = 2$ and $L = 4$ with 30/30 paired wins.
- The final implementation is forward-exact and, in the tested code path, reduces step time and peak memory.

Supported with qualification

- Shallow $L = 1$ models improve on average under RSTOPALL, but individual seeds can degrade substantially; the strongest claim should therefore remain depth-dependent.
- SGD showed a weak positive trend but was floor-limited because the baseline deep QORLA model rarely trained under the predeclared SGD settings. Non-Adam transfer is instead supported by RMSprop.
- STAB2-33E is a provisional blind task-preservation pass because the interrupted run lost complete update-tail bookkeeping for nine of ten seeds.

Rejected hypotheses

- The Stability Test 1 conditioning threshold is universal in trainable models.
- Mass-normalized conditioning or leverage alone provides a robust universal event trigger.
- Fixed or EMA-normalized correction dominance provides a universal branch cap.
- Scalar raw-gradient shrinkage necessarily implies optimizer-step shrinkage under Adam.
- Large optimizer-space dominance D_A is intrinsically harmful in real supervised training.
- Current-minibatch descent alignment is sufficient for safe deep-stack intervention.
- A single early $\|R\|/\|B\|$ predictor generalizes across unseen trajectories.

18 Limitations and next stage

Stability Test 2 is complete for the investigated small-model regime, but it does not establish a theorem that correction stop-gradient is optimal for every QORLA architecture. The main limitations are: the supervised dataset is small; the deepest validated model has four QORLA blocks; multi-head QORLA, long sequences, large embedding widths, mixed precision, distributed training, and high-resolution vision or language benchmarks have not yet been tested; and the final rule deliberately sacrifices direct learning through R , raising a representation-capacity question that must be studied at scale.

The next campaign is therefore Stability Test 3: end-to-end scaling and comparative stability. Its primary axes are sequence length, width, depth, multi-head construction, FP32/BF16/FP16 precision, long-horizon optimization, memory/runtime scaling, and direct comparisons with softmax attention and efficient-attention baselines.

STABILITY TEST 2 IS FINISHED.

The retained deep-model mechanism is forward-exact correction stop-gradient: keep $o = B + R$ numerically unchanged, but prevent gradients from propagating through R during training.

A Experiment chronology

Experiment	Focus	Main result	Verdict
STAB2-01A	Healthy trainable Q/K/V	Damping silent; forward/loss/gradients transparent in healthy geometry.	PASS
STAB2-01B	Stressed trainable Q/K/V	Condition-triggered intervention activates with high gradient-direction fidelity.	PASS
STAB2-02	Learned geometry	Permanent damping traps geometry through conditioning persistence.	PARTIAL
STAB2-03	Hard release	Geometry recovers but abrupt release causes rebound Q-gradient spike.	FAIL
STAB2-04	Smooth release	Linear threshold release controls rebound while preserving loss.	PASS
STAB2-05	Five-seed confirmation	Strong Q/K spike reduction; V-gradient caveat.	PASS
STAB2-06–09	Sequence length	Old conditioning score changes with N under matched geometry.	DIAG.
STAB2-10–13	Mass normalization	κ_{mass} repairs controlled N invariance, but thresholds still fail learned dynamics.	PARTIAL
STAB2-14–19	Leverage/event prediction	Moderate predictive association but weak precision/recall for intervention.	REJECT
STAB2-20	Exact branch decomposition	Correction dominance strongly tracks key-gradient spikes; no major cancellation.	STRONG PASS
STAB2-21–22	Branch-relative cap	Strong synthetic suppression with small loss change.	PASS
STAB2-23–27	Full-block transfer	Mechanism transfers, fixed threshold does not.	PARTIAL
STAB2-28	Relative-EMA cap	Fails universal robustness, especially short sequence.	FAIL
STAB2-29–30	Power compression	Threshold-free whole-branch compression suppresses gradients but is nearly always active.	PARTIAL
STAB2-31	Held-out power rule	Raw whole-branch rule degrades task on held-out seeds.	FAIL
STAB2-31B	Relative-power EMA	No robust improvement.	FAIL
STAB2-31C	K-only gradient surgery	Strong raw-gradient reduction, uncertain task effect.	PARTIAL
STAB2-31D	Adam audit	Adam almost cancels scalar gradient suppression at update level.	KEY FAIL
STAB2-31E	PCD development	Adam-preconditioned correction control genuinely suppresses updates.	STRONG PASS
STAB2-31F	Blind PCD	50-run optimizer-tail confirmation; tiny-loss relative artifacts at $N = 32$.	STRONG PASS
STAB2-31G	$N = 32$ convergence	Useful tolerance convergence approximately identical.	PASS
STAB2-31H	Full-horizon vanishing p	No sufficient ordinary-Adam recovery period; late convergence worsens.	FAIL
STAB2-31I/J	Finite transient window	30% transient window balances early stabilization and tail convergence.	STRONG PASS
STAB2-31K/L	Frozen blind TPCD + stats	50/50 update-tail wins; cluster-bootstrap confirms effect.	STRONG PASS
STAB2-32A	Matched teacher	Validation parity/improvement and 25/25 update-tail wins.	STRONG PASS
STAB2-32B	Mismatched teacher	25/25 validation wins and strong update suppression.	STRONG PASS
STAB2-33A	Real digits, stressed init	Optimizer stability transfers; task preservation mixed.	PARTIAL
STAB2-33B	Real digits, normal init	Dominance-only TPCD harms task despite smaller updates.	FAIL/PARTIAL
STAB2-33C	Correction utility	Most large- D_A corrections are locally useful; D_A not a harm detector.	STRONG DIAG.
STAB2-33D	DP-TPCD development	Selective intervention repairs task preservation; active only about 6%.	STRONG PASS
STAB2-33E	DP-TPCD blind	Task side passes; update-tail aggregate incomplete after interrupted runtime.	PROVISIONAL PASS
STAB2-34A	Depth generalization	DP-TPCD fails increasingly with depth.	FAIL
STAB2-34B	Descent projection	Minimal projection still hurts task and tails.	FAIL
STAB2-34C	Layer localization	Every individually projected block can hurt.	FAIL

Experiment	Focus	Main result	Verdict
STAB2-34D	Cross-batch utility	39–44% of current-batch “harmful” labels reverse on probe batch.	DIAG. PASS
STAB2-34E	Cross-batch consensus	Tail improvement but severe trajectory bifurcation.	FAIL
STAB2-34F	Batch-order sensitivity	Baseline itself has large trajectory variance.	STRONG DIAG.
STAB2-34G	Layerwise amplification	Development data suggest early block-0 amplification.	PARTIAL DIAG.
STAB2-34H	Early warning window	Block-0 R/B appears predictive on development trajectories.	DIAG. PASS
STAB2-34I	Blind predictor test	R/B predictor fails completely on unseen trajectories.	FAIL
STAB2-34J	Causal branch ablation	KSTOP ineffective; ALLSTOP wins 20/20 and rescues deep collapse.	STRONG CAUSAL
STAB2-34K	Depth confirmation	Forward equality zero; RSTOPALL strong at $L = 2, 4$, variable at $L = 1$.	STRONG PASS
STAB2-35A	Adam vs AdamW	RSTOPALL: 19/20 and 20/20 wins, no collapse runs.	STRONG PASS
STAB2-35B	SGD	Baseline floor-limited; weak positive RSTOPALL trend.	INCONCLUSIVE
STAB2-35C	RMSprop	20/20 wins; large accuracy gains; non-Adam transfer.	STRONG PASS
STAB2-36A	Backward dose response	Monotonic accuracy/loss improvement as correction gradient scale goes to zero.	VERY STRONG
STAB2-36B	Generic baselines	RSTOPALL far exceeds low LR and clipping; 15/15 wins.	VERY STRONG
STAB2-37A	Frozen blind multi-split	42/45 wins overall; 30/30 deep wins; strong Wilcoxon significance.	VERY STRONG
STAB2-38A	Numerical/runtime audit	Forward error 0; invalid steps 0; faster and lower-memory backward.	FINAL PASS

B Key late-stage numerical tables

B.1 STAB2-34K: depth confirmation

Depth	BASE acc.	RSTOP30 acc.	RSTOPALL acc.	RSTOPALL wins
1	62.11	77.00	71.50	3/5
2	24.06	55.67	74.28	5/5
4	10.50	18.94	66.17	5/5

B.2 STAB2-35: optimizer transfer

Optimizer	BASE acc.	RSTOPALL acc.	Mean gain	Wins
Adam	13.44	55.65	+42.21 pp	19/20
AdamW	12.57	57.33	+44.76 pp	20/20
RMSprop 10^{-3}	12.61	48.36	+35.75 pp	10/10
RMSprop 3×10^{-3}	15.83	54.94	+39.11 pp	10/10

B.3 STAB2-36A: correction-backward dose response

α	Mean test loss	Mean accuracy	Collapses < 20%
1.00	2.3077	10.67	15/15
0.50	2.2570	14.06	13/15
0.25	1.8392	27.41	5/15
0.00	1.0939	59.69	0/15

B.4 STAB2-36B: generic stabilization comparison

Method	Mean test loss	Mean accuracy	Accuracy gain	Collapse count
BASE	2.2470	12.61	–	13/15
Global clip 1.0	2.1633	16.15	+3.54 pp	9/15
Low LR 10^{-3}	2.1236	16.72	+4.11 pp	10/15
RSTOPALL	1.0134	62.65	+50.04 pp	0/15

B.5 STAB2-38A: computational audit

Depth	Time ratio	Memory ratio	BASE grad	RSTOP grad	BASE bad	RSTOP bad
1	0.65694	0.97031	10.9096	0.6542	0	0
2	0.73363	0.94837	2.1659	0.6021	0	0
4	0.71327	0.91358	0.6642	0.4167	0	0

C Compact mathematical summary of the discarded and retained rules**Stability Test 1 conditioning damping**

$$s = \min\left(1, \frac{\text{sg}(\kappa)}{0.03}\right), \quad \tilde{R} = \text{sg}(R) + s(R - \text{sg}(R)). \quad (44)$$

Useful at operator level, but not universal under trainable geometry.

Preconditioned correction dominance

$$D_A = \frac{\|u_R\|}{\|u_B\| + \epsilon}, \quad s = \min(1, D_A^{-p}), \quad \Delta W_K = -\eta(u_B + su_R). \quad (45)$$

Strong synthetic optimizer-step stabilization; not universally task-preserving on normal real data.

Descent-preserving TPCD

$$s = \begin{cases} 1, & g_K^\top u_R > 0, \\ \min(1, D_A^{-p}), & g_K^\top u_R \leq 0. \end{cases} \quad (46)$$

Selective and effective for a one-block real task, but fails depth generalization.

Final retained rule

$$\boxed{\tilde{R} = \text{sg}(R), \quad \text{value}(\tilde{R}) = R, \quad \frac{\partial \tilde{R}}{\partial \theta} = 0.} \quad (47)$$

The forward QORLA operator remains exactly $B + R$; only the correction backward path is removed.